Dual implementation pricing: the city's charges from the social optimum, with one best response per carrier per round

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Theory notes, direction C

Abstract

The city (leader) chooses per-delivery and per-vehicle-hour charges for N carriers who play the exposure-based (D1) Cournot–congestion game of `theory/main.tex`, with plans from an NP-hard routing oracle. The project's current leader method is a grid over two charge parameters with one equilibrium solve per point (about 2,400 oracle calls per city). We replace the search by an implementation step. (i) Because every lot's vehicle-hours are proportional to its deliveries, the two instruments collapse to one number u_z per zone, and a uniform per-delivery charge plus non-negative zone congestion charges realise every $u \in \mathbb{R}^Z$ (Proposition 1). (ii) An exact identity links the carriers' potential under u to the social cost (Lemma 2): with the tangent charge $u_z = \kappa_z Q_z^\circ - \frac{m_z}{2}\bar{d}_z$ (a Pigou term minus a lumpy Cournot subsidy) the two differ by a concave quadratic in the zone aggregates and a linear lot-asymmetry term. Consequences: the tangent charge implements the convexified social optimum exactly when lots are symmetric, and every convexified equilibrium is then socially optimal (Theorem 5); with asymmetric lots the welfare loss is at most $(1+\rho)E$, with E an explicit lot-asymmetry number (Theorem 4). (iii) In the pure game the charges that make a given target profile an equilibrium form a polyhedron whose constraints are separated by *one best response per carrier at the target*; a cutting-plane LP finds such a charge or proves that none exists, in finitely many rounds of N oracle calls and with no equilibrium solve (Theorem 7). When it succeeds, the leader attains the social cost of the target, so the only remaining loss is the suboptimality of the social solver (Proposition 8). (iv) Under Γ-maximin imprecision of width δ the implementing set is translated by the surcharge, so the charges for every δ follow from one run (Theorem 12). The individual ingredients (Pigou–Cournot corrective taxes, toll polyhedra, inverse optimisation by cutting planes) are classical; Section 8 says exactly what is new. Experiments (`results.md`) show the method on nine real city games with the exact branch-and-price oracle.

1 Setting

Assumption 1 (City game; `src/ccu/zmc_brl.py`)**.** Zones $z \in [Z]$, carriers $i \in [N]$. Carrier i has customer zones $C_i \subseteq [Z]$ with lot sizes $d_{iz} > 0$ ($z \in C_i$; put $d_{iz} = 0$ otherwise) and a finite plan set $\mathcal{P}_i$. A plan p serves a set $S(p) \subseteq C_i$ (all-or-nothing lots) with routes of routing cost $R_i(p) \geq 0$ (the BRL upper expectation). Deliveries $q_{iz}(p) = d_{iz}\mathbf{1}[z \in S(p)]$ and vehicle-hours $\tau_{iz}(p) = r\, q_{iz}(p)$ for one ratio $r > 0$ (`h = d * h_drop`). Inverse demand $\alpha_z - b_z Q_z$, congestion cost $\gamma c_z X_z$ per vehicle-hour ($g_z := \gamma c_z$), with $Q = \sum_i q_i$, $X = \sum_i \tau_i = rQ$. The city posts s_z per delivery and t_z

per vehicle-hour. Carrier i's cost at the pure profile σ is

$$J_i(p;\sigma_{-i}) = R_i(p) + \sum_z \Big[q_{iz}\big(-\alpha_z + b_z(q_{iz}+Q_{-i,z}) + s_z \big) + \tau_{iz}\big(g_z(\tau_{iz}+X_{-i,z}) + t_z \big) \Big].$$

The social cost (charges are transfers) is $\mathrm{SC}(\sigma) = \sum_i R_i(\sigma_i) + \sum_z \big[-\alpha_z Q_z + \frac{b_z}{2} Q_z^2 + g_z X_z^2 \big]$.

Write $\kappa_z = g_z r^2$ (congestion slope in delivery units), $m_z = b_z + \kappa_z$ (the slope the carriers face) and $m_z^S = b_z + 2\kappa_z$ (the social slope). Since $q_{iz} \in \{0, d_{iz}\}$ we have $q_{iz}^2 = d_{iz} q_{iz}$, which is used repeatedly.

Proposition 1 (One instrument per zone)**.** [PROVED] *(a) For every plan, $\sum_z (s_z q_{iz} + t_z \tau_{iz}) = \langle u, q_i \rangle$ with $u_z = s_z + r t_z$. Hence the game depends on (s,t) only through $u \in \mathbb{R}^Z$. (b) Every $u \in \mathbb{R}^Z$ is realised by a* uniform *per-delivery charge $s = \min_{z \in Z_+} u_z$ (a subsidy if negative) and non-negative zone congestion charges $t_z = (u_z - s)/r$, where Z_+ is the set of zones with at least one lot. (c) The grid of* `experiments/run_leader.py` *uses $s_z \equiv s$ and $t_z = \theta\, g_z X_z^{\mathrm{NE}}$, i.e. the two-dimensional family $u = s\mathbf{1} + \theta\, r\, g \odot X^{\mathrm{NE}}$.*

Proof. (a) $\tau_{iz} = r q_{iz}$. (b) $s + r t_z = u_z$ and $t_z \geq 0$ because $s \leq u_z$; zones without lots carry no plan and are irrelevant. (c) Substitution. □

With u as the instrument, $J_i^u(p;\sigma_{-i}) = R_i(p) + \sum_z q_{iz}\big(-\alpha_z + m_z(d_{iz} + Q_{-i,z}) + u_z \big)$ and the game has the exact potential (D1; `main.tex`, Theorem 4.4, Monderer–Shapley)

$$\Phi_u(\sigma) = \sum_i R_i(\sigma_i) + \sum_z \Big[-\alpha_z Q_z + \tfrac{m_z}{2}\big(Q_z^2 + \textstyle\sum_i q_{iz}^2 \big) + u_z Q_z \Big],$$

as one checks directly: changing σ_i from p to p' changes Φ_u by $J_i^u(p') - J_i^u(p)$.

2 The potential–welfare identity

Lemma 2 (Identity)**.** [PROVED] *Fix $Q^\circ \in \mathbb{R}^Z$ and $\bar{d} \in \mathbb{R}^Z$ and let $u_z^{\tan} = \kappa_z Q_z^\circ - \frac{m_z}{2}\bar{d}_z$. For every u put $v = u - u^{\tan}$. Then for every pure profile σ*

$$\Phi_u(\sigma) = \mathrm{SC}(\sigma) - \sum_z \tfrac{\kappa_z}{2}(Q_z - Q_z^\circ)^2 + L(\sigma) + \langle v, Q \rangle + \sum_z \tfrac{\kappa_z}{2}(Q_z^\circ)^2, \qquad L(\sigma) = \sum_{i,z} \tfrac{m_z}{2}(d_{iz} - \bar{d}_z) q_{iz}. \quad (1)$$

Proof. With $q_{iz}^2 = d_{iz} q_{iz}$, $X_z = r Q_z$ and $m_z^S - m_z = \kappa_z$, $\Phi_u - \mathrm{SC} = \sum_z \big[\frac{m_z}{2} \sum_i d_{iz} q_{iz} - \frac{\kappa_z}{2} Q_z^2 + u_z Q_z \big]$. Insert $u_z = \kappa_z Q_z^\circ - \frac{m_z}{2}\bar{d}_z + v_z$ and complete the square: $-\frac{\kappa_z}{2} Q_z^2 + \kappa_z Q_z^\circ Q_z = -\frac{\kappa_z}{2}(Q_z - Q_z^\circ)^2 + \frac{\kappa_z}{2}(Q_z^\circ)^2$, and $\frac{m_z}{2}\sum_i d_{iz} q_{iz} - \frac{m_z}{2}\bar{d}_z Q_z = \sum_i \frac{m_z}{2}(d_{iz} - \bar{d}_z) q_{iz}$. □

The tangent charge is the classical corrective tax of a Cournot market with an externality (marginal external damage $\kappa_z Q_z$ minus a market-power subsidy; Buchanan 1969, Barnett 1980). Two details are specific to the lumpy lots: the subsidy is $\frac{m_z}{2} d$ per delivery (half a lot, and it includes the carriers' *own* congestion κ_z), not $b_z q_i$; and the externality is evaluated at the target aggregate, not at the unpriced equilibrium. In ZMC price terms (`theory/new_algorithm_A`, $\lambda = MA$), $\kappa_z Q_z^\circ = \lambda_z^{SO} - \lambda_z$ is the gap between the social and the carriers' zone price *at the same aggregate.*

Proposition 3 (Personalised charges; classical)**.** [CLASSICAL] *(marginal-contribution utilities, Wolpert–Tumer; Marden–Wierman 2013). Let σ^* be a pure profile and $U_{iz} = \kappa_z Q^*_{-i,z} - \frac{b_z}{2} d_{iz}$. Under carrier-specific per-delivery charges U_i, for all i and $p, p' \in \mathcal{P}_i$: $J_i^U(p; \sigma^*_{-i}) - J_i^U(p'; \sigma^*_{-i}) = \mathrm{SC}(p, \sigma^*_{-i}) - \mathrm{SC}(p', \sigma^*_{-i})$. Hence every profile that no single carrier can improve socially (in particular every social optimum) is a pure equilibrium.*

Proof. $\mathrm{SC}(p, \sigma^*_{-i}) = R_i(p) + \sum_z q_{iz}\big(-\alpha_z + m_z^S Q^*_{-i,z} + \frac{m_z^S}{2} d_{iz}\big) + \text{const}$, and $J_i^U(p) = R_i(p) + \sum_z q_{iz}\big(-\alpha_z + m_z d_{iz} + m_z Q^*_{-i,z} + U_{iz}\big)$; the brackets coincide because $m_z + \kappa_z = m_z^S$ and $m_z - \frac{b_z}{2} = \frac{m_z^S}{2}$. □

A city cannot charge per carrier. Flat charges reproduce U_{iz} only up to the spread $\kappa_z d_{iz} + \frac{b_z}{2}(d_{iz} - d_{jz})$ across carriers; the rest of the paper measures what this costs and how to avoid it.

3 The convexified game

Let each carrier mix, $x_i \in \Delta(\mathcal{P}_i)$, and write $\bar{q}_i = \mathbb{E}_{x_i} q_i$, $\bar{Q} = \sum_i \bar{q}_i$. The convexified potential (the relaxation of `main.tex` Section 5 and of ZMC) and the convexified social cost are

$$\overline{\Phi}_u(x) = \sum_i \mathbb{E}_{x_i}\Big[R_i + \sum_z q_{iz}\big(-\alpha_z + u_z + \tfrac{m_z}{2} d_{iz}\big)\Big] + \sum_z \tfrac{m_z}{2} \bar{Q}_z^2, \qquad \overline{\mathrm{SC}}(x) = \sum_i \mathbb{E}_{x_i}\Big[R_i - \sum_z \alpha_z q_{iz}\Big] + \sum_z \tfrac{m_z^S}{2} \bar{Q}_z^2.$$

Both are convex on the product of simplices, they agree with Φ_u and SC up to the own-term, which is linear in x because $q_{iz}^2 = d_{iz} q_{iz}$, and the minimisers of $\overline{\Phi}_u$ are the convexified (relaxed) equilibria. Let x^* minimise $\overline{\mathrm{SC}}$, with aggregate $\bar{Q}^*$. The identity of Lemma 2 holds verbatim for $\overline{\Phi}_u$ and $\overline{\mathrm{SC}}$ with Q, q replaced by $\bar{Q}, \bar{q}$ (same algebra; both sides are affine in x except the quadratic terms in $\bar{Q}$).

Theorem 4 (Asymmetric lots)**.** [PROVED] *Let $u = \kappa \odot \bar{Q}^* - \frac{m}{2} \odot \bar{d}$ for any $\bar{d}$. Every convexified equilibrium x^u (minimiser of $\overline{\Phi}_u$) satisfies*

$$0 \le \overline{\mathrm{SC}}(x^u) - \overline{\mathrm{SC}}(x^*) \le (1+\rho)\,\big[L(x^*) - L(x^u)\big] \le (1+\rho)\,E, \qquad \rho = \max_z \frac{\kappa_z}{b_z + \kappa_z} < 1,$$

with $L(x) = \sum_{i,z} \frac{m_z}{2}(d_{iz} - \bar{d}_z)\bar{q}_{iz}$ and the lot-asymmetry number $E = \sum_{i,z} \frac{m_z}{2} |d_{iz} - \bar{d}_z|\, d_{iz}$. Moreover $\sum_z \frac{b_z + \kappa_z}{2}(\bar{Q}_z^u - \bar{Q}_z^*)^2 \le L(x^*) - L(x^u)$.

Proof. $\overline{\mathrm{SC}}$ is a linear function of x plus $\sum_z \frac{m_z^S}{2} \bar{Q}_z^2$ with $\bar{Q}$ linear in x, so the second-order expansion is exact: $\overline{\mathrm{SC}}(x) - \overline{\mathrm{SC}}(x^*) = \langle \nabla \overline{\mathrm{SC}}(x^*), x - x^* \rangle + \sum_z \frac{m_z^S}{2}(\bar{Q}_z - \bar{Q}_z^*)^2$, and the inner product is non-negative by optimality of x^* over the convex feasible set. Hence $\Delta := \overline{\mathrm{SC}}(x^u) - \overline{\mathrm{SC}}(x^*) \ge \sum_z \frac{m_z^S}{2}(\Delta \bar{Q}_z)^2$. By the identity with $Q^\circ = \bar{Q}^*$, $v = 0$: $\overline{\Phi}_u(x) = \overline{\mathrm{SC}}(x) - \sum_z \frac{\kappa_z}{2}(\bar{Q}_z - \bar{Q}_z^*)^2 + L(x) + \text{const}$. Since x^u minimises $\overline{\Phi}_u$, $\overline{\Phi}_u(x^u) \le \overline{\Phi}_u(x^*)$, i.e. $\Delta \le \sum_z \frac{\kappa_z}{2}(\Delta \bar{Q}_z)^2 + L(x^*) - L(x^u)$. Combining the two bounds, $\sum_z \frac{m_z^S - \kappa_z}{2}(\Delta \bar{Q}_z)^2 \le L(x^*) - L(x^u)$, which is the last claim since $m_z^S - \kappa_z = b_z + \kappa_z$. Then $\sum_z \frac{\kappa_z}{2}(\Delta \bar{Q}_z)^2 \le \rho \sum_z \frac{b_z + \kappa_z}{2}(\Delta \bar{Q}_z)^2 \le \rho\,[L(x^*) - L(x^u)]$ and $\Delta \le (1+\rho)[L(x^*) - L(x^u)]$. Finally $|L(x^*) - L(x^u)| \le \sum_{i,z} \frac{m_z}{2} |d_{iz} - \bar{d}_z|\, |\bar{q}^*_{iz} - \bar{q}^u_{iz}| \le E$ because $\bar{q}_{iz} \in [0, d_{iz}]$. □

Theorem 5 (Exact implementation of the convexified optimum)**.** [PROVED] *If lots are symmetric within zones ($d_{iz} = d_z$ for all i with $z \in C_i$), the tangent charge $u_z = \kappa_z \bar{Q}_z^* - \frac{m_z}{2} d_z$ makes x^* a convexified equilibrium, and* every *convexified equilibrium under u has aggregate $\bar{Q}^*$ and social cost $\overline{\mathrm{SC}}(x^*)$.*

Algorithm 1 DIP: dual implementation pricing

Target. Column generation on $\overline{\mathrm{SC}}$ (the social-cost variant of the carriers' oracle), rounding, coordinate polish $\Rightarrow$ pure σ^*, aggregate Q^*.
Reference. $u^{\mathrm{ref}} = \kappa \odot Q^* - \frac{m}{2} \odot \bar{d}$ (tangent charge, $\bar{d}_z$ = mean lot in z); cuts $\mathcal{C} \leftarrow \emptyset$; $u \leftarrow u^{\mathrm{ref}}$.
for $k = 1, 2, \ldots$ **do**
  for $i = 1..N$ **do** $b_i \leftarrow$ oracle best response to σ^*_{-i} under u ▷ N routing calls
    if $J_i^u(b_i) < J_i^u(\sigma_i^*)$ **then** add the cut $\langle q_i(b_i) - q_i(\sigma_i^*), u \rangle \geq J_i^0(\sigma_i^*) - J_i^0(b_i) + \varepsilon$ to $\mathcal{C}$
  if no cut was added **then return** u, σ^* ▷ σ^* is an exact pure equilibrium under u
  $u \leftarrow \arg\min\{\|u - u^{\mathrm{ref}}\|_1 : \ u \in \mathcal{U},\ u \text{ satisfies } \mathcal{C}\}$ (LP); **if** infeasible: **break**
Fallback. Best-response dynamics from σ^* under the least-violating LP charge and under u^{ref}; keep the better equilibrium.
Instruments. $s = \min_z u_z$, $t_z = (u_z - s)/r$ (Proposition 1).

Proof. With $\bar{d} = d$, L is identically zero on the feasible set (each term has $d_{iz} = \bar{d}_z$ or $\bar{q}_{iz} = 0$), so $E = 0$ in Theorem 4: every minimiser x^u of $\overline{\Phi}_u$ has $\Delta\bar{Q} = 0$ and $\overline{\mathrm{SC}}(x^u) = \overline{\mathrm{SC}}(x^*)$. It remains to show that x^* itself minimises $\overline{\Phi}_u$: $\overline{\Phi}_u = \overline{\mathrm{SC}} - \sum_z \frac{\kappa_z}{2}(\bar{Q}_z - \bar{Q}_z^*)^2 + \text{const}$, the subtracted term has zero gradient at x^*, so $\nabla\overline{\Phi}_u(x^*) = \nabla\overline{\mathrm{SC}}(x^*)$ and the first-order optimality condition of the convex function $\overline{\Phi}_u$ (curvature $b_z + \kappa_z > 0$ in $\bar{Q}_z$) holds at x^*. □

Remark 6 (What the user's first idea gets right and wrong). The implementing charge is indeed "social zone price minus carriers' zone price, minus a market-power wedge", but (i) the two prices must be evaluated at the *same* aggregate (the target), not at the unpriced equilibrium; (ii) the wedge is $\frac{m_z}{2} d_{iz}$, half a lot including own congestion, and not $b_z Q_z/N$; (iii) in the convexified game no refinement step is needed at all under symmetric lots (Theorem 5). The refinement is needed only for lot asymmetry and for integrality, and the right tool for it is not a secant on the charge (each secant step needs an equilibrium solve, about 40 oracle calls) but the cutting plane of Section 4, whose rounds cost one best response per carrier.

4 The pure game: implementation polyhedron and cutting planes

Let $\mathcal{U} = \prod_z [\underline{u}_z, \overline{u}_z]$ be the admissible charges (in the code $[-\alpha_z, \alpha_z]$). For a pure target σ^* let J_i^0 denote J_i at $u = 0$ and define

$$P(\sigma^*) = \big\{ u \in \mathcal{U} : \ \langle q_i(p) - q_i(\sigma_i^*), u \rangle \ \geq \ J_i^0(\sigma_i^*; \sigma_{-i}^*) - J_i^0(p; \sigma_{-i}^*) \quad \forall i,\ \forall p \in \mathcal{P}_i \big\}.$$

By Proposition 1(a), $J_i^u = J_i^0 + \langle u, q_i \rangle$, so $P(\sigma^*)$ is exactly the set of admissible charges under which σ^* is a pure Nash equilibrium. It is a polyhedron (finitely many constraints), possibly empty. This "toll polyhedron" viewpoint is classical for nonatomic routing (Bergendorff–Hearn–Ramana 1997; Hearn–Ramana 1998) and for inverse optimisation (Ahuja–Orlin 2001).

Theorem 7 (Separation and finite termination). [PROVED] *(a) For $u \in \mathcal{U}$, $u \in P(\sigma^*)$ if and only if no carrier has a strictly improving best response to σ^*_{-i} under u; if carrier i's best response b_i improves, the constraint (i, b_i) is violated by u, and it is the most violated constraint of carrier i (largest $J_i^u(\sigma_i^*) - J_i^u(p)$). (b) With $\varepsilon = 0$, Algorithm 1 stops after at most $1 + \sum_i (|\mathcal{P}_i| - 1)$ rounds, either with $u \in P(\sigma^*)$ or with an infeasible LP, and the latter happens only if $P(\sigma^*) = \emptyset$. With $\varepsilon > 0$ the same holds for the shrunken polyhedron $P_\varepsilon(\sigma^*)$, whose elements make σ^* a strict equilibrium against every plan with a different delivery vector. Each round costs N oracle calls and*

no equilibrium solve. (c) [CLASSICAL] *If the LP step is replaced by a central cut (centre of gravity or ellipsoid, Grötschel–Lovász–Schrijver) and $P_\varepsilon(\sigma^*)$ contains a ball of radius ϱ in the k coordinates of the zones that appear in some deviation, a point of $P_\varepsilon(\sigma^*)$ is found in $O(k\log(D/\varrho))$ rounds (centre of gravity) or $O(k^2\log(D/\varrho))$ rounds (ellipsoid), D the diameter of $\mathcal{U}$.*

Proof. (a) $J_i^u(p) = J_i^0(p) + \langle u, q_i(p)\rangle$, so constraint (i,p) reads $J_i^u(p) \geq J_i^u(\sigma_i^*)$; all of carrier i's constraints hold iff $\min_p J_i^u(p) \geq J_i^u(\sigma_i^*)$, and the violation of (i,p) is $J_i^u(\sigma_i^*) - J_i^u(p)$, maximised by a best response. (b) If round k adds a cut (i,b_i), the current u_k violates it; u_k satisfies every earlier cut (it is LP-feasible), so the cut is new. There are $\sum_i(|\mathcal{P}_i|-1)$ non-trivial constraints, which bounds the number of rounds that add cuts. Every cut is a valid constraint of $P(\sigma^*)$, so an infeasible LP proves emptiness. The statement for $\varepsilon > 0$ is the same argument applied to the right-hand sides increased by ε. (c) Standard: each round returns either a point of the set or a separating halfspace through the current centre; the volume of the localiser shrinks by a constant factor (centre of gravity: $1 - 1/e$; ellipsoid: $e^{-1/(2(k+1))}$) and cannot fall below the volume of the ball. □

Proposition 8 (Single-level form and optimality)**.** [PROVED] *Let $V^\star = \inf_{u\in\mathcal{U}} \min_{\sigma\in\mathrm{NE}(u)} \mathrm{SC}(\sigma)$ be the leader's value with optimistic selection. Then $V^\star = \min\{\mathrm{SC}(\sigma) : \sigma \text{ pure}, P(\sigma)\neq\emptyset\} \geq \min_\sigma \mathrm{SC}(\sigma) \geq \min_x \overline{\mathrm{SC}}(x)$. If Algorithm 1 ends in its main loop, it returns charges under which σ^* is an equilibrium, so $\mathrm{SC}(\sigma^*) - V^\star \leq \mathrm{SC}(\sigma^*) - \min_\sigma \mathrm{SC}(\sigma)$: the leader's loss is only the suboptimality of the social solver. In particular it is at least as good as any grid over any subfamily of $\mathcal{U}$.*

Proof. $\sigma \in \mathrm{NE}(u)$ iff $u \in P(\sigma)$, so the two descriptions of $V^\star$ range over the same pairs (u,σ). The other inequalities are inclusions of feasible sets ($\min \overline{\mathrm{SC}}$ relaxes pure profiles to mixtures, and $\overline{\mathrm{SC}} = \mathrm{SC}$ at pure profiles because $q^2 = dq$). □

The leader problem is therefore a single-level problem over profiles with an implementability side condition; DIP solves its relaxation (drop the side condition) and checks the side condition at the relaxed optimum with the carriers' own oracle. Selection is the remaining issue: other equilibria may exist under the returned u (Remark 11).

Theorem 9 (Loss of the fallback; a posteriori and a priori)**.** [PROVED] *Let u be any charge, $v = u - u^{\tan}$ with $u^{\tan} = \kappa \odot Q^* - \frac{m}{2} \odot \bar{d}$, and let σ^u be reached from σ^* by strictly improving unilateral moves under u (best-response dynamics; it terminates in a pure equilibrium because Φ_u strictly decreases on a finite set). Then*

$$\mathrm{SC}(\sigma^u) - \mathrm{SC}(\sigma^*) \;\leq\; \sum_z \tfrac{\kappa_z}{2}(Q_z^u - Q_z^*)^2 + L(\sigma^*) - L(\sigma^u) + \langle v, Q^* - Q^u\rangle$$

$$\leq\; \sum_z \tfrac{\kappa_z}{2}\left(\max\{Q_z^*,\; Q_z^{\mathrm{full}} - Q_z^*\}\right)^2 + E + \textstyle\sum_z |v_z|\, Q_z^{\mathrm{full}},$$

with $Q_z^{\mathrm{full}} = \sum_i d_{iz}$. For $u = u^{\tan}$ and symmetric lots only the congestion term remains.

Proof. $\Phi_u(\sigma^u) \leq \Phi_u(\sigma^*)$; substitute (1) with $Q^\circ = Q^*$ on both sides; the constant cancels and $Q^* - Q^* = 0$ on the right. The a priori bound uses $0 \leq Q_z^u \leq Q_z^{\mathrm{full}}$, the bound on $|L(\sigma^*) - L(\sigma^u)|$ from the proof of Theorem 4, and $|Q_z^* - Q_z^u| \leq Q_z^{\mathrm{full}}$. □

Corollary 10 (Oracle calls)**.** [PROVED] *DIP uses $C_{\mathrm{SO}} + N\,R + C_{\mathrm{BR}}$ oracle calls: C_{SO} for the social target (column generation, at most N calls per iteration plus a coordinate polish), R cutting-plane rounds (bounded in Theorem 7) and, only if the polyhedron is empty, C_{BR} calls of best-response*

dynamics from σ^* (at most N per strict improvement of Φ_u, plus a final verifying pass). No step solves an equilibrium from scratch. A grid with G points and an equilibrium solve of C_{eq} calls per point uses GC_{eq}.

Remark 11 (Selection). DIP returns the target as an *exact* pure equilibrium under the returned charges (verified by best responses, and in the experiments by a fresh-pool oracle). Other equilibria may coexist, as in the grid method, whose equilibria are those selected by ZMC from warm starts. Two tools reduce the risk: the margin $\varepsilon > 0$ (strictness), and the identity (1): under the tangent charge with symmetric lots, an equilibrium σ' that is a global minimiser of Φ_u satisfies $\mathrm{SC}(\sigma') \leq \mathrm{SC}(\sigma^*) + \sum_z \frac{\kappa_z}{2}(Q'_z - Q^*_z)^2$, so the potential-minimising equilibrium can lose at most a congestion-sized term. The experiments report, in addition, the equilibrium that the grid's own protocol (ZMC from the unpriced prices) selects under DIP's charges.

5 Credal imprecision

Theorem 12 (Translation by the surcharge)**.** [PROVED] *Under Γ-maximin credal boxes of width δ and weights w (`main.tex`, Lemma 3.6, Assumption 2.4), the Γ-maximin equilibria under u are the pure equilibria of the game with the per-unit surcharge $\mathrm{sur}(\delta) = \delta\, Mw \in \mathbb{R}^{2Z}$ added to the contribution. With $\hat{s}_z(\delta) = \delta(b_z w_{(z,Q)} + r\, g_z w_{(z,X)})$:*

$$P^{\Gamma}_{\delta}(\sigma^*) = P(\sigma^*) - \hat{s}(\delta) \qquad (\textit{intersected with } \mathcal{U}).$$

Hence one run of DIP gives the implementing charges $u(\delta) = u(0) - \hat{s}(\delta)$ for every $\delta \geq 0$ at no further oracle cost, and as long as $u(\delta) \in \mathcal{U}$ the leader attains the same social cost $\mathrm{SC}(\sigma^*)$ at every width: the welfare loss of ambiguity aversion that E4 measures without charges is entirely removable by charges. For the criteria E-admissibility, maximality and interval dominance, $u(0)$ itself makes σ^* an imprecise equilibrium at every δ, because $\mathrm{ICNE}_C(0) = \mathrm{NE} \subseteq \mathrm{ICNE}_C(\delta)$ (`main.tex`, Lemma 3.3).

Proof. By Lemma 3.6 of `main.tex` under D1, carrier i's Γ-maximin payoff is its precise payoff minus $\delta\langle a_i(p), Mw\rangle$ (traffic enters the BRL route loss already at its upper expectation). With $a_i = (q_i, rq_i)$, $\delta\langle a_i(p), Mw\rangle = \langle \hat{s}(\delta), q_i(p)\rangle$, so $J_i^{u,\Gamma} = J_i^{u+\hat{s}(\delta)}$ and $u \in P^{\Gamma}_{\delta}(\sigma^*) \iff u + \hat{s}(\delta) \in P(\sigma^*)$. The social cost does not contain the surcharge (it is a belief, not a resource cost). The last statement is the monotonicity lemma. □

6 Two instruments

Proposition 13 (When a uniform per-delivery charge suffices)**.** [PROVED] *(a) The pair (uniform s, zone $t \geq 0$) realises every zone charge (Proposition 1); a zone-specific per-delivery charge adds nothing to it, and a per-vehicle-hour charge adds nothing to a zone-specific per-delivery charge. The only thing the congestion instrument contributes beyond a uniform s is spatial differentiation. (b) Under symmetric lots, a uniform charge implements the convexified optimum exactly (in the sense of Theorem 5) if the tangent charge $\kappa_z \bar{Q}^*_z - \frac{m_z}{2} d_z$ is the same number in every active zone. The condition is sufficient, not necessary: zones at a vertex of the optimum tolerate one-sided deviations, which the cutting plane of Section 4 exploits. (c) With the calibration of `build_city_game`, $b_z = \alpha_z/(2Q^{\mathrm{full}}_z)$ and $\alpha_z = 1.5$ fee. If the n_z carriers present in zone z have equal lots, $Q^{\mathrm{full}}_z = n_z d_z$ and the congestion-free tangent charge is $u_z = -\frac{b_z}{2} d_z = -\frac{\alpha_z}{4n_z} = -\frac{0.375}{n_z}$ fee: a subsidy of 37.5% of the fee in single-carrier zones and 19% in two-carrier zones. A uniform subsidy is exact only if every*

served zone has the same number of carriers; otherwise the zone congestion charge is needed to tax zones with more competitors relative to the others.

Proof. (a) Proposition 1. (b) Theorem 5, since the tangent charge is then uniform. (c) Substitution, with $m_z \approx b_z$ when $\kappa_z \ll b_z$. □

The grid's best points (`docs/EXPERIMENTS.md`, E2: $s/\text{fee} \in [-0.4, -0.2]$) sit exactly in the range predicted by (c), which explains why the per-delivery subsidy captures most of the gain and why the congestion charge helps where the number of competitors varies across zones.

7 Experiments in brief

Details in `results.md`. Exact branch-and-price oracle, nine cities, seeds 0–2 (27 runs). In the three cities with a complete 63-point exact grid (London, Buenos Aires, New Delhi), DIP reached 104–135% of the grid-optimal social-cost gain with 4.9–9.4% of the grid's oracle calls; in all 27 runs it reached at least 0.999 of the best grid point with at most 9.4% of the calls of a 63-point grid. The target was implemented exactly in all 27 runs (the tangent charge alone sufficed in 18; at most 3 cutting-plane rounds), so the priced equilibrium attains the social cost of the column-generation optimum. On 210 enumerated tiny games, DIP hit the exact leader optimum over all zone charges in 93% of the games, and in 100% when given the exact social optimum as target. The zone charges follow the rule $u_z \approx -0.375\,\text{fee}/n_z$ of Proposition 13(c) (correlation 0.98–1.00).

8 Prior art and what is new

- **Corrective taxes under oligopoly** (Buchanan 1969; Barnett 1980; Benchekroun–Long 1998 for dynamic Cournot): the optimal per-unit tax is the marginal external damage minus a market-power correction. The tangent charge is this rule; the half-lot, own-congestion form is a small adaptation to all-or-nothing lots. *Classical.*

- **Marginal-contribution / Pigouvian personalised charges** (Wolpert–Tumer; Marden–Wierman 2013): Proposition 3. *Classical.*

- **Toll polyhedra and implementing a target flow** (Bergendorff, Hearn, Ramana 1997; Hearn–Ramana 1998; Fotakis–Spirakis 2008 cost-balancing tolls for atomic games; Bhaskar, Ligett, Schulman, Swamy, FOCS 2014 / GEB 2019, ellipsoid with an *equilibrium* oracle; Stoll 2021 for unique equilibria in atomic unsplittable games). *Classical framework.* Bhaskar et al. query an equilibrium flow for each candidate toll, i.e. one equilibrium solve per query; here the query is one best response per carrier at the fixed target, which is why the method needs no equilibrium solve.

- **Inverse optimisation by cutting planes** (Ahuja–Orlin 2001; Wang 2009 for inverse MILP; Bodur–Chan–Zhu 2022): Algorithm 1's loop is an inverse-optimisation cutting plane, one inverse problem per carrier coupled through the common charge. *Classical technique.*

- **Leader search with an equilibrium solver in the loop** (Masiha, Elahi, Kiyavash, Thiran, arXiv 2602.23277, 2026: zeroth-order outer updates of tolls over a Frank–Wolfe equilibrium solver in combinatorial congestion games, including freight carriers with NP-hard tour problems). This is the closest leader-side neighbour; like the project's grid, every outer step needs an equilibrium solve.

- **Pure equilibria in integer-programming games** (Dragotto–Scatamacchia, Zero Regrets, IJOC 2023; Carvalho et al. Cut-and-Play; Duguet–Harks–Schmidt–Schwarz 2025): cutting planes over profiles with best-response separation, without a leader or charges.

What we did not find and claim: (1) the exact potential–welfare identity (1) for the lumpy Cournot–congestion game, and the loss bounds it gives (Theorems 4, 5, 9) with the observable lot-asymmetry number E; (2) the combination "social target by the same column-generation machinery + implementation polyhedron separated by the carriers' own NP-hard best responses at the target" as an algorithm for the leader of an atomic aggregative routing game, with the single-level characterisation of Proposition 8; (3) the translation of the implementing set by the Γ-maximin surcharge (Theorem 12); (4) the competitor-count rule of Proposition 13(c). These are modest; the contribution is mainly that the leader problem of this project needs one social solve and a few best-response rounds instead of a grid of equilibrium solves.

Limits. (i) The quality of the result is the quality of the social target; the column-generation target is a heuristic pure optimum (Proposition 8 makes this the only loss when the polyhedron is non-empty). (ii) Optimistic selection: the target is an exact equilibrium under the returned charges, but not necessarily the only one. (iii) The structure used is specific: D1, one hours-per-delivery ratio r (collinearity), affine inverse demand and linear congestion. Without collinearity the per-zone instrument is two-dimensional and Proposition 1(b) fails; the polyhedron and the cutting plane still work in $\mathbb{R}^{2Z}$. (iv) If $P(\sigma^*) = \emptyset$ the fallback is a heuristic with the a posteriori bound of Theorem 9; no rate is claimed for retargeting.